Submodule 35.15: Branch Commands & Management

1. Overview of Branching Operations

Git provides a comprehensive command suite to inspect, create, switch, rename, and delete branches safely.

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2. Essential Branch Commands

1. Listing All Branches:

```
git branch
```

- Lists all local branches in the repository.
- An asterisk (`*`) and green text indicates your currently active (checked out) branch.

```
* main
  feature-auth
  bugfix-dataset
```

2. Creating a New Branch:

```
git branch <branch_name>
```

Example:

```
git branch feature-login
```

(Creates the branch pointer at the current commit without switching to it).

3. Switching Between Branches (`checkout` vs `switch`):

```
# Classic command
git checkout feature-login
```

```
# Modern Git command (Git 2.23+)
git switch feature-login
```

- Updates files in your working directory to match the snapshot of the target branch.
- Moves the `HEAD` pointer to `feature-login`.

4. Create and Switch in One Step:

```
# Classic shortcut
git checkout -b feature-login

# Modern shortcut
git switch -c feature-login
```

5. Renaming a Branch:

```
# Rename currently active branch
git branch -M main

# Rename specific branch
git branch -m old_name new_name
```

6. Deleting a Branch:

```
# Safe deletion (aborts if branch has unmerged commits)
git branch -d feature-login

# Force deletion (deletes branch regardless of merge status)
git branch -D feature-login
```

> [!CAUTION] You cannot delete a branch that you are currently standing on. Always checkout to another branch (e.g. `main`) before executing `git branch -d`.